Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	9100Spark
Project Name	Import Data using Transform Maps
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the architectural diagram below and the information as per Table-1 and Table-2.

Architecture: bulk employee data from a spreadsheet is loaded into ServiceNow, staged in an Import Set table, transformed into the target table using a Transform Map, then validated and reported.

Step	Data flowin thearchitecture
1	The administrator prepares the employee data in Google Sheets and downloads it as Sample Spreadsheet.xlsx
2	Through the browser, the file is uploaded with System Import Sets → Load Data into the Import Set table Employee Import (u_employee_import), which acts as a staging table
3	The Transform Map Sample Spreadsheet Import maps each source field to the Employee Test (u_employee_test) target table; Coalesce is enabled on Employee ID
4	Running the transform inserts new employees and updates existing ones; the run is recorded in Transform History
5	Reports (Employees by Department, Employees by Location, Employee List) read the target table and are shown on the Employee Analytics Dashboard
6	Roles, ACLs and sharing settings control who can load data, create dashboards and view the reports

Architecture Diagram:

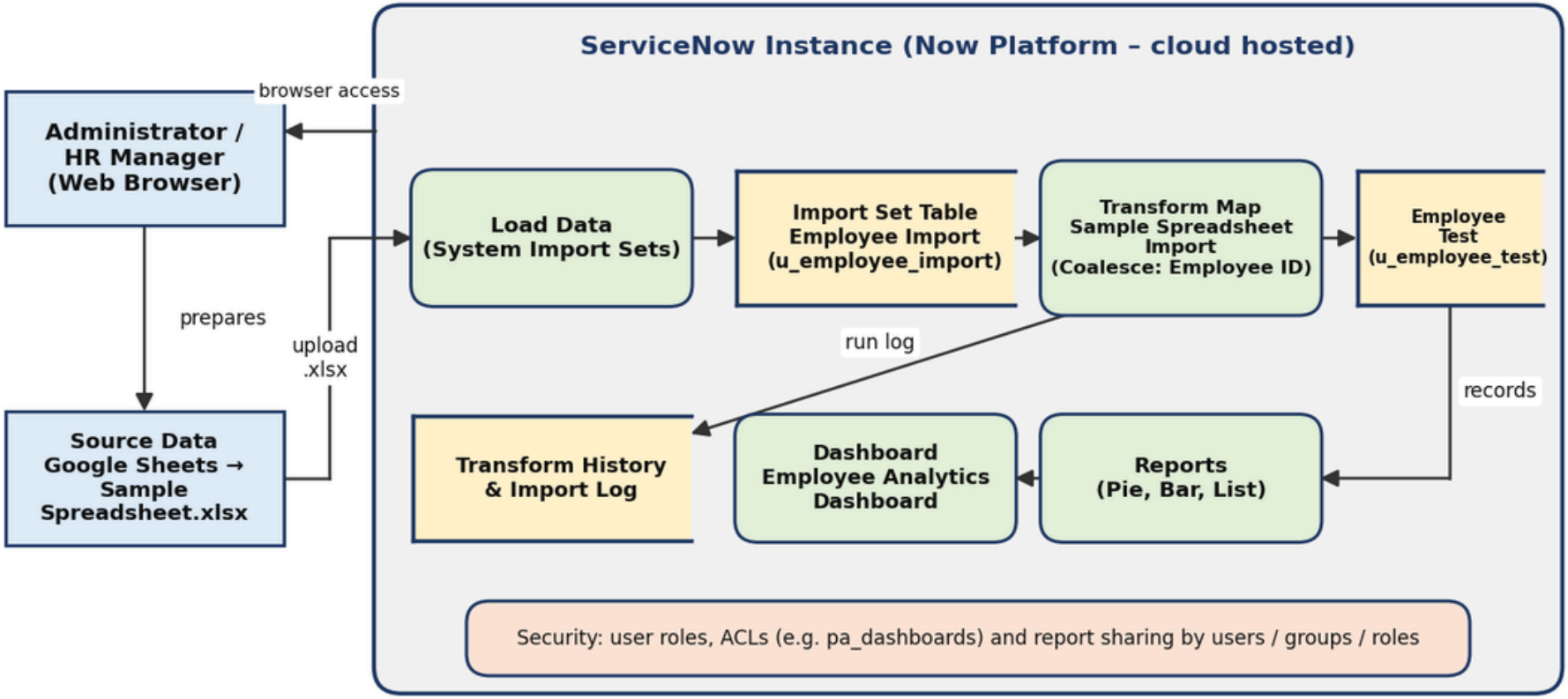


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How the user interacts with the application: web UI with Application Navigator, forms and lists	ServiceNow Platform UI (web browser)
2	Application Logic-1	Data loading: upload the spreadsheet and create the Import Set (staging) table	ServiceNow System Import Sets – Load Data
3	Application Logic-2	Field mapping and transformation of staged data into the target table	ServiceNow Transform Map (Auto Map Matching Fields, field maps)
4	Application Logic-3	Duplicate prevention: update existing records and insert only new ones	Coalesce on Employee ID (Transform Map field map)
5	Database	Custom tables: u_employee_import (staging) and u_employee_test (target); fields Employee ID, Employee Name, Email, Department, Location as String	ServiceNow platform tables
6	Cloud Database	Database service that stores the tables and records on the instance	ServiceNow cloud instance database (platform-managed)
7	File Storage	Source spreadsheet file used for the import	Sample Spreadsheet.xlsx (Google Sheets export) on the local file system, uploaded through Load Data
8	External API-1	Not used – the import is file based	N/A
9	External API-2	Not used	N/A
10	Machine Learning Model	Not used	N/A
11	Infrastructure (Server / Cloud)	Application deployment: cloud-hosted ServiceNow instance accessed through a browser; no local server is required	ServiceNow Now Platform (cloud)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	None – the solution is configured on the ServiceNow low-code platform	ServiceNow platform configuration (Import Sets, Transform Maps, Reports, Dashboards)
2	Security Implementations	Role-based access to load data and manage dashboards; dashboard creation is controlled by the pa_dashboards ACL (Admin overrides); report and dashboard sharing restricted by users, groups and roles	ServiceNow roles and Access Control Lists (ACLs)
3	Scalable Architecture	A staging (Import Set) table is separated from the target table, so the same Transform Map can be reused for larger files and new spreadsheets	Import Set table + Transform Map on a cloud platform
4	Availability	The application is available through the cloud-hosted ServiceNow instance from any browser; imports can be re-run using the same Transform Map	ServiceNow cloud instance
5	Performance	Bulk loading and a single transform run replace manual entry; Transform History reports the total, inserted, updated and ignored rows for each run	Import Set, Transform Map, Transform History